SCSI1113- COMPUTATIONAL MATHEMATICS
2021-2022/ Semester 1
Assignment 4 Linear Transformation

1. Determine whether $T: M_{2,2} \to R, T(A) = a + b + c + d,$ where $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is a linear transformation.

2. Let $T: \mathcal{R}^3 \to \mathcal{R}^3$ is a linear transformation defined by , $T(v_1, v_2, v_3) = (v_2 - v_1, v_1 + v_2, 2v_1)$

   a) Find image of $v = (2,3,0)$
   b) Find preimage of $w = (-11, -1, 10)$

3. Let $T: \mathcal{R}^2 \to \mathcal{R}^2$ be a linear transformation, and let $B = (v_1, v_2)$ is a basis in $\mathcal{R}^3$,
   If $v_1 = \begin{bmatrix} 2 \\ 3 \end{bmatrix}, T\left(\begin{bmatrix} 2 \\ 3 \end{bmatrix}\right) = \begin{bmatrix} 8 \\ 1 \end{bmatrix}$, $v_2 = \begin{bmatrix} 4 \\ 1 \end{bmatrix}, T\left(\begin{bmatrix} 4 \\ 1 \end{bmatrix}\right) = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$

   Find $T\left(\begin{bmatrix} 10 \\ 5 \end{bmatrix}\right)$

4. Let $T: \mathcal{R}^5 \to \mathcal{R}^3$ is a linear transformation defined by $T(v) = Av$, where,

$$A = \begin{bmatrix} -1 & 3 & 2 & 1 & 4 \\ 2 & 3 & 5 & 1 & 4 \\ 2 & 0 & 2 & 2 & 0 \end{bmatrix}$$

   Find a basis for the null space of *T* and its dimension

5. Let $T: \mathcal{R}^3 \to \mathcal{R}^3$ be a linear operator and $B = \{v_1, v_2, v_3\}$ a basis of $\mathcal{R}^3$. Suppose

$$T(v_1) = \begin{bmatrix} -1 \\ 2 \\ 1 \end{bmatrix} \quad T(v_2) = \begin{bmatrix} 0 \\ 5 \\ 0 \end{bmatrix} \quad T(v_3) = \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix}$$

   a) Determine whether $u = \begin{bmatrix} -2 \\ 1 \\ 2 \end{bmatrix}$ is in range of *T*
   b) Find a basis for *R(T)*

6. Let $T: \mathcal{R}^3 \to \mathcal{R}^4$ is a linear transformation defined by

$$T\left(\begin{bmatrix} x \\ y \\ z \end{bmatrix}\right) = \begin{bmatrix} 2x \\ x + y \\ y + z \\ x + z \end{bmatrix}$$

basis $B = \left\{\begin{bmatrix} 2 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ 1 \end{bmatrix}\right\}$ $B' = \left\{\begin{bmatrix} 1 \\ 0 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 0 \\ 0 \end{bmatrix}\right\}$, ordered basis for $\mathcal{R}^3$ and $\mathcal{R}^4$,

   a) Find matrix relative to $[T]_B^B$,
   b) Let $v = \begin{bmatrix} 1 \\ -5 \\ 2 \end{bmatrix}$, find T(v) using matrix found in(a)